

INTERVIEW PREP SERIES

.NET Developer Interview Questions

130+ real questions from recent interviews — sorted into 14 topics, no company names attached.

SWIPE / WATCH THROUGH →

01

TOPIC 01 / 14

SQL & Database

21 questions in this topic

1

Schema change on a live database with data integrity and minimal downtime

2

ROW_NUMBER vs RANK vs DENSE_RANK vs NTILE

3

Deadlock from delete+reinsert vs a UI read — why it happens and how to resolve it

4

Using a read replica or read-only table to avoid deadlocks

5

Why a deadlock happens in production but not in UAT

6

Lock types on delete (row/page/table-level) — few rows vs thousands of rows

7

DB-side performance checks; analyzing query performance and execution plans

8

SQL transactions — why they're used, ACID, and Durability

9

Foreign key vs manually storing an ID

10

Indexes in SQL

11

How to identify missed records during a migration

12

Getting the 2nd of 5 weekly snapshots

13 Finding the 2nd highest salary per department

14 Prerequisites to create a table

15 Primary Key vs Unique Key

16 CTE vs Temp Table

17 How to analyze a query execution plan

18 Stored Procedure vs Function — can a function call another function?

19 What is a Composite Key

20 Can a non-clustered index be created on a Primary Key?

21 SQL Triggers



02

TOPIC 02 / 14

C# & .NET Fundamentals

19 questions in this topic



1 What is the CLR, and what does it do besides execution?

2 Garbage Collection

3 Compilation, and Compile-time vs Run-time errors

4 Error handling in .NET

5 Value Type vs Reference Type, with an object/reference output puzzle

6 Constructors — how many, overloading, why use multiple, what goes inside

7 Boxing and Unboxing

8 Why a static class needs no object

9 List vs IEnumerable

10 Functions vs Methods

11 What is a Class

12 IList vs IEnumerable

13 Delegates; Func, Action, and Predicate

14 var vs int

15 ref vs out

16 Can a non-static class have static methods? Can a static class have only static members?

17 Can a constructor be private?

18 The base keyword in inheritance

19

readonly vs const



03

TOPIC 03 / 14

Dependency Injection & Object Lifetimes

7 questions in this topic



1 Controlling object lifetime in the backend

2 Singleton, Scoped, and Transient explained

3 What is Dependency Injection, and why not just use "new"?

4 Types of Dependency Injection

5 AddScoped vs AddTransient

6 Designing an EmployeeService (CRUD) — which lifetime, and why not Singleton or Transient

7

What is IServiceCollection

04

TOPIC 04 / 14

Async / Await & Threading

8 questions in this topic

1

Feature-flag cache: thread-safety of an in-memory dictionary, race conditions during background sync

2

Task vs Thread; thread starvation

3

Async/Await basics, real use cases, the benefit for a single DB call

4

Is Async/Await the same as multithreading? How do they free up threads?

5

What is a Task in .NET

6

Mixing async and sync methods in one orchestrator — what happens if you skip await?

7

What await actually does internally; do dependent async calls wait or move on?

8

What is a Race Condition

05

TOPIC 05 / 14

Entity Framework Core

10 questions in this topic

1

Managing tables with EF, Code First vs DB First, and the Fluent API

2

Promoting schema changes across environments

3

How migrations work, and how EF tracks which migrations are applied

4

Task.WhenAll with parallel EF calls, and the DbContext thread-safety issue

5

Sharing the same DbContext across 2 async methods and 1 sync method — what breaks

6

AsNoTracking — when to use it, and what happens if you modify and SaveChanges anyway

7 Configuring Lazy Loading

8 One-to-Many / Many-to-One relationships

9 What is an EDMX file

10 Executing a stored procedure via EF

06

TOPIC 06 / 14

Middleware & Request Pipeline

7 questions in this topic

1

What is Middleware, and a walkthrough of the HTTP request pipeline

2

Designing global exception handling

3

`app.Use()` vs `app.Run()`

4

Custom Middleware

5

The ASP.NET Core pipeline end-to-end; adding a global custom header

6

`app.Use` vs `app.Run`, plus Attribute vs Conventional Routing

7

How the Exception Handler Middleware works internally

07

TOPIC 07 / 14

API Security, Auth & JWT

10 questions in this topic

1

Managing authentication/authorization; blocking non-admin users from an admin API

2

Securing API endpoints; Rate Limiting; JWT creation and structure; Refresh Tokens

3

CORS — the concept, a real implementation, and when to update it

4

Throttling

5

CORS specifically in production, and resolving it after a URL change

6

Cookies vs Session/Local Storage security

7

Detecting token expiry and triggering a refresh

8

If JWT.io can regenerate a token, how does validation actually work?

9

API security best practices; SQL Injection; JWT ID Token vs Access Token

10

Cookie lifetime



08

TOPIC 08 / 14

Azure, Cloud & Diagnostics

10 questions in this topic



1

Deploying a full-stack app on Azure for scalability and reliability

2

Monitoring and diagnosing performance on Azure (App Service + auto-scale)

3

Diagnosing sudden API response-time spikes with App Insights

4

Investigating sudden production performance degradation

5

Memory leaks in ASP.NET Core — what they are and how they happen

6

Designing enterprise logging and monitoring on Azure

7

Azure Functions

8

SQL is fast but the API is slow — how do you troubleshoot?

9

Moving data from RDS to Azure Blob Storage

10

Logging API response time, and the value of logging beyond error tracking



09

TOPIC 09 / 14

Microservices & Architecture

3 questions in this topic

1

What is a Microservice

2

Monolithic vs Microservices

3

Inter-service communication, Content Negotiation, and changing the response format

10

TOPIC 10 / 14

React & Frontend

5 questions in this topic

1

What is a Component; passing data from parent to child

2

State management in large React apps; Context API vs Redux; fixing re-render slowness

3

React Hooks; Custom Hooks; when to reach for Redux

4

Angular lifecycle hooks

5

Native apps vs React Native



11

TOPIC 11 / 14

Design Patterns & Code Reviews

8 questions in this topic



1

Code Review: why calling `.Result` on async code is a problem, and the fix

2

Code Review: spotting the N+1 Query Problem and how to fix it

3

Code Review: reviewing and documenting a snippet, with suggested improvements

4

Interfaces vs Abstract Classes

5

Liskov Substitution Principle

6

Builder Pattern; Unit of Work Pattern

7

Encapsulation, Abstraction, and the four pillars of OOP

8

What is a ViewModel for

12

TOPIC 12 / 14

System Design & Scenarios

2 questions in this topic

1

A file upload crash — diagnosis, fix, and server-side validation

2

Designing a Feature Flag System with an in-memory cache and sync store

13

TOPIC 13 / 14

Behavioral & Communication

2 questions in this topic

1

Adapting your communication style across Agile roles

2

Tell me about yourself, and your current project/tech stack



14

TOPIC 14 / 14

Testing, Web Basics & AI

11 questions in this topic



1 Serialization in .NET, with an example

2 Unit Testing; hitting 80% coverage fast; AI tools for test generation

3 GET vs POST — why separate verbs?

4 What is LTM?

5 ChromaDB / vector DBs; AI vs Generative AI vs Agent AI vs Agentic AI

6 List vs Set vs Tuple in Python

7

CSS Box Model; HTML Elements; JS let vs var vs const

8

Disconnected Architecture in ADO.NET; executing a stored procedure via ADO.NET

9

Hangfire

10

ModelState validation — what it is and why you check IsValid

11

FizzBuzz without using continue



That's all 14 topics.

130+ questions — SQL, C#/.NET, EF Core, Async, Security, Azure, React, Design Patterns & more.

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